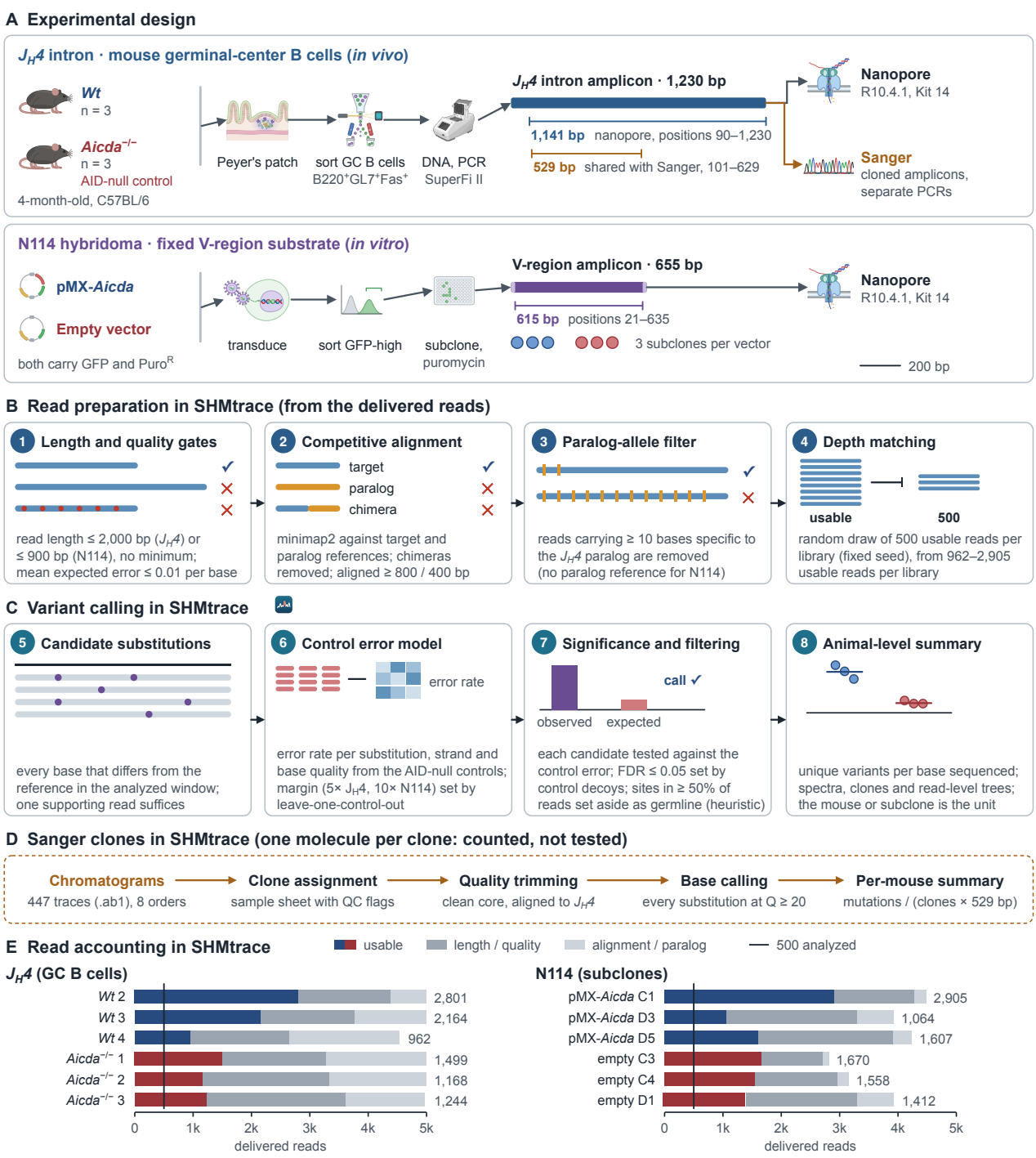

A Experimental design
JH4 intron · mouse germinal-center B cells (in vivo)
Wt
n = 3
Aicda−/−
n = 3
AID-null control
4-month-old, C57BL/6
Peyer's patch
sort GC B cells
B220+GL7+Fas+
DNA, PCR
SuperFi II
JH4 intron amplicon · 1,230 bp
1,141 bp nanopore, positions 90–1,230
529 bp shared with Sanger, 101–629
Nanopore
R10.4.1, Kit 14
Sanger
cloned amplicons,
separate PCRs
N114 hybridoma · fixed V-region substrate (in vitro)
pMX-Aicda
Empty vector
both carry GFP and PuroR
transduce
sort GFP-high
subclone,
puromycin
V-region amplicon · 655 bp
615 bp positions 21–635
3 subclones per vector
Nanopore
R10.4.1, Kit 14
200 bp
B Read preparation in SHMtrace (from the delivered reads)
1 Length and quality gates
read length ≤ 2,000 bp (JH4) or
≤ 900 bp (N114), no minimum;
mean expected error ≤ 0.01 per base
2 Competitive alignment
target
paralog
chimera
minimap2 against target and
paralog references; chimeras
removed; aligned ≥ 800 / 400 bp
3 Paralog-allele filter
reads carrying ≥ 10 bases specific to
the JH4 paralog are removed
(no paralog reference for N114)
4 Depth matching
usable
500
random draw of 500 usable reads per
library (fixed seed), from 962–2,905
usable reads per library
C Variant calling in SHMtrace
5 Candidate substitutions
every base that differs from the
reference in the analyzed window;
one supporting read suffices
6 Control error model
error rate
error rate per substitution, strand and
base quality from the AID-null controls;
margin (5× JH4, 10× N114) set by
leave-one-control-out
7 Significance and filtering
call ✓
observed
expected
each candidate tested against the
control error; FDR ≤ 0.05 set by
control decoys; sites in ≥ 50% of
reads set aside as germline (heuristic)
8 Animal-level summary
unique variants per base sequenced;
spectra, clones and read-level trees;
the mouse or subclone is the unit
D Sanger clones in SHMtrace (one molecule per clone: counted, not tested)
Chromatograms
447 traces (.ab1), 8 orders
Clone assignment
sample sheet with QC flags
Quality trimming
clean core, aligned to JH4
Base calling
every substitution at Q ≥ 20
Per-mouse summary
mutations / (clones × 529 bp)
E Read accounting in SHMtrace
usable
length / quality
alignment / paralog
500 analyzed
JH4 (GC B cells)
Wt 2
Wt 3
Wt 4
Aicda−/− 1
Aicda−/− 2
Aicda−/− 3
2,801
2,164
962
1,499
1,168
1,244
0
1k
2k
3k
4k
5k
delivered reads
N114 (subclones)
pMX-Aicda C1
pMX-Aicda D3
pMX-Aicda D5
empty C3
empty C4
empty D1
2,905
1,064
1,607
1,670
1,558
1,412
0
1k
2k
3k
4k
5k
delivered reads